Kevin Berry

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SKILLS

Python, Rust, Go, Bash, C, JavaScript, Linux, Kernel, Git, Machine Learning, Statistics, Containers, etc.

EXPERIENCE

Software Engineer - Container Optimized OS (COS) Kernel Jan 2024 - Present
Google *Sunnyvale, CA*

- Led kernel development for an open-source Linux distro used by key Google Cloud products like GKE and Cloud SQL
- Responsible for keeping the kernel updated, secure, and reliable for all users
- Led major kernel upgrades, coordinating with many teams and external companies to add and verify driver support
- Tested drivers and integrated them into the COS kernel (Lustre, iRDMA, TPU, IDPF, and others)
- Made a package manager for kernel modules (cos-dkms); reduced time to release out-of-tree modules from weeks to days
- Led investigations which resulted in order-of-magnitude speedups for file transfer workloads on COS
- Triaged and solved customer issues relating to many major Linux subsystems like file systems, networking, storage, etc.
- Triaged and patched thousands of kernel and package vulnerabilities
- Led a Linux kernel reading group to increase the skills and knowledge of the whole team

Machine Learning Engineer Jun 2017 - Dec 2023
Worthix - Research Team *Alpharetta, GA*

- Developed LLM-based API for large scale multi-level feedback analysis and summarization
- Developed and trained NLP models and API for real-time survey response topic classification
- Developed novel algorithms for large-scale unsupervised graph classification
- Developed queries and statistical analyses for client-facing dashboards
- Deployed and monitored APIs and applications as scalable microservices in Azure
- Conducted scientific experiments to improve survey data and model quality
- Developed fast Monte-Carlo Shapley value approximation for calculating response topic importance
- Developed reverse-geolocation API capable of resolving millions of point-in-region queries per second
- Contributed Python code to networkx which massively improved performance for graph unions and intersections
- Contributed Rust code to polars dataframe library which solved infinite loop in core groupby operations

Teaching Assistant May 2016 - Dec 2017
Computer Organization and Programming (CS 2110) *Atlanta, GA*

- Led recitations on C, Assembly, CPU datapaths, and digital logic
- Wrote software to automate grading of Java programs and circuits

SELECTED PROJECTS

image-to-ascii (Rust): Uses computer vision techniques to convert images and gifs to ASCII art. Capable of converting 120+ images per second. Uses hand-coded SIMD instructions for improved performance. (Github) (Crates.io)

detect-duplicates (Rust): Finds all of the duplicated files in a directory. Reads each file at most twice and, in expectation, only has one file loaded at a time. Can check the entire Linux kernel tree in about 5 seconds. (Github) (Crates.io)

Readability Analyzer (JavaScript): Assigns average text grade level based on standard readability metrics. Uses MLP with hand-engineered features to count syllables. Can analyze 2,000,000+ characters per second. (Web App)

EDUCATION

Georgia Institute of Technology, B.S. Computer Science Aug 2015 - Dec 2018
Selected Coursework: Machine Learning, Linguistics, Computer Graphics, Computer Vision, Compilers, Linear Algebra
GPA: 3.96/4.0